

25 Supersymmetry Algebras

25.5 Massive Particle Supermultiplets

Although the known quarks, leptons, and gauge bosons and their superpartners may probably be treated as massless at energies where supersymmetry breaking is negligible, this is not necessarily true for other particles, including the extra gauge bosons of large mass required by theories that unify the strong and electroweak interactions. Also, ever since the Wess–Zumino model, massive particle theories have been useful test cases for studying supersymmetry theories. It will therefore be worthwhile for us briefly to consider the implications of unbroken supersymmetry for massive particles.

As in the previous section, we obtain the various one-particle states in a supermultiplet by acting on any one of them with the operators Q_{ar} and $Q_{ar}^\dagger$, and all of these states have the same four-momentum. Unlike the case of zero mass, for mass $M > 0$ we may now take this to be the four-momentum of a particle at rest, with $p^i = 0$ for $i = 1, 2, 3$ and $p^0 = M$. In this frame of reference, we have

$$\sigma_\mu p^\mu = M\sigma_0 = M \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (25.5.1)$$

Thus, acting on any state $\| \rangle$ in a supermultiplet with this four-momentum, the anticommutation relation (25.2.7) yields

$$\{Q_{ar}, Q_{bs}^\dagger\} \| \rangle = 2M\delta_{ab}\delta_{rs} \| \rangle. \quad (25.5.2)$$

In contrast with the case of zero mass, here no component of Q_{ar} or $Q_{ar}^\dagger$ can vanish on the whole multiplet, so we have two sets of raising and lowering operators: both $Q_{(1/2)r}$ and $Q_{(-1/2)r}^\dagger$ lower the spin 3-component by $1/2$, while both $Q_{(-1/2)r}$ and $Q_{(1/2)r}^\dagger$ raise the spin 3-component by $1/2$. However, as we shall see, for extended supersymmetry it is possible for certain linear combinations of the Q s and $Q^\dagger$ s to vanish.

We will first consider the case of simple supersymmetry. By using the supersymmetry algebra (25.2.31), (25.2.32), we shall show that the general massive supermultiplet consists of a particle of spin $j+1/2$, a pair of particles of spin j , and a particle of spin $j-1/2$. Where parity is conserved, the particles of spin $j \pm 1/2$ have equal intrinsic parity, given by some phase η , while the two particles of spin j have parities $+i\eta$ and $-i\eta$. Here j is any integer or half-integer greater than zero. There is also a collapsed supermultiplet, consisting of two particles of spin zero and a particle of spin $1/2$. When parity is conserved the particles of spin zero have parities $i\eta$ and $-i\eta$, where η is the parity of the particle of spin $1/2$.

Here is the proof. We first show that any supermultiplet will contain at least one spin multiplet of states $\|j, \sigma\rangle$ with spin 3-component σ running by unit steps from $-j$ to $+j$, having the special property that, for all such σ and for $a = \pm 1/2$,

$$Q_a \|j, \sigma\rangle = 0. \quad (25.5.3)$$

Starting with any non-zero state $\|\psi\rangle$ in the supermultiplet, we can define non-zero states

$$\|\psi'\rangle \equiv \begin{cases} (2M)^{-1/2} Q_{1/2} \|\psi\rangle, & Q_{1/2} \|\psi\rangle \neq 0, \\ \|\psi\rangle, & Q_{1/2} \|\psi\rangle = 0, \end{cases}$$

and

$$\|\psi''\rangle \equiv \begin{cases} (2M)^{-1/2} Q_{-1/2} \|\psi'\rangle, & Q_{-1/2} \|\psi'\rangle \neq 0, \\ \|\psi'\rangle, & Q_{-1/2} \|\psi'\rangle = 0. \end{cases}$$

Because the Q_a anticommute, $Q_{1/2} \|\psi'\rangle = 0$, and so $Q_a \|\psi''\rangle = 0$ for $a = \pm 1/2$. If any state $\|\psi''\rangle$ satisfies the condition that $Q_a \|\psi''\rangle = 0$, then so does $U(R) \|\psi''\rangle$, where $U(R)$ is the unitary operator representing an arbitrary spatial rotation. It follows that the states that satisfy this condition may be decomposed into complete spin multiplets $\|j, \sigma\rangle$, satisfying condition (25.5.3).

Now focus on any one of these spin multiplets satisfying Eq. (25.5.3), normalized so that

$$\langle j, \sigma' \| j, \sigma \rangle = \delta_{\sigma' \sigma}. \quad (25.5.4)$$

For $j > 0$, by applying the spin 1/2 operators¹ $Q_a^\dagger$ to these states we can construct states of spin $j \pm 1/2$:

$$\|j \pm 1/2, \sigma\rangle = \frac{1}{\sqrt{2M}} \sum_a C_{\frac{1}{2}j}(j \pm 1/2, \sigma; a, \sigma - a) Q_a^\dagger \|j, \sigma - a\rangle, \quad (25.5.5)$$

where $C_{jj'}(j'', \sigma''; \sigma, \sigma')$ is the conventional Clebsch–Gordan coefficient for coupling spins j and j' with 3-components σ and σ' to make spin j'' with 3-component σ'' . Using Eqs. (25.5.2)–(25.5.5) and the orthonormality properties of the Clebsch–Gordan coefficients, we can show that these states are properly normalized:

$$\langle j \pm 1/2, \sigma \| j \pm 1/2, \sigma' \rangle = \delta_{\sigma \sigma'}, \quad \langle j \pm 1/2, \sigma \| j \mp 1/2, \sigma' \rangle = 0, \quad (25.5.6)$$

so none of the states $\|j \pm 1/2, \sigma\rangle$ can vanish. The only exception is for $j = 0$, in which case of course there is no state $\|j - 1/2, \sigma\rangle$. We can also construct other states by applying *two* $Q_a^\dagger$ s to $\|j, \sigma\rangle$. Since each $Q_a^\dagger$ anticommutes with itself, the only such non-zero states are formed by applying the operator $Q_{1/2}^\dagger Q_{-1/2}^\dagger = -Q_{-1/2}^\dagger Q_{1/2}^\dagger$. This operator may be written as $\frac{1}{2} e_{ab} Q_a^\dagger Q_b^\dagger$, which shows that it is rotationally invariant, so this gives a second spin multiplet with spin j :

$$\|j, \sigma\rangle^b = \frac{1}{2M} Q_{1/2}^\dagger Q_{-1/2}^\dagger \|j, \sigma\rangle, \quad (25.5.7)$$

which is distinguished from $\|j, \sigma\rangle$ by the fact that instead of Eq. (25.5.3), we have

$$Q_a^\dagger \|j, \sigma\rangle^b = 0. \quad (25.5.8)$$

Again using Eqs. (25.5.2)–(25.5.4), we find that these are also normalized states:

$${}^b\langle j, \sigma' \| j, \sigma \rangle^b = \delta_{\sigma' \sigma}, \quad \langle j, \sigma' \| j, \sigma \rangle^b = 0. \quad (25.5.9)$$

¹Since Q_a transforms under rotations like a field that destroys a particle of spin 1/2 and spin 3-component a , it is $Q_a^\dagger$ that transforms like a field that creates such a particle, and hence transforms like the particle itself. To put this formally, $[J_i, Q_a] = -\sum_b \frac{1}{2} (\sigma_i)_{ab} Q_b$, so $[J_i, Q_a^\dagger] = \sum_b \frac{1}{2} (\sigma_i)_{ba} Q_b^\dagger$, which may be compared with the transformation property of a spin 1/2 particle, $J_i \|a\rangle = \sum_b \frac{1}{2} (\sigma_i)_{ba} \|b\rangle$.

It is then easy to show that the states constructed so far form a complete representation of the supersymmetry algebra. The orthonormality property of the Clebsch–Gordan coefficients allows us to rewrite Eq. (25.5.5) as

$$Q_a^\dagger \|j, \sigma\rangle = \sqrt{2M} \sum_{\pm} C_{\frac{1}{2}j}(j \pm 1/2, \sigma + a; a, \sigma) \|j \pm 1/2, \sigma + a\rangle. \quad (25.5.10)$$

Also, Eq. (25.5.2) shows that, for any state $\| \rangle$ in the supermultiplet,

$$[Q_a, Q_{1/2}^\dagger Q_{-1/2}^\dagger] \| \rangle = 2M \sum_b e_{ab} Q_b^\dagger \| \rangle, \quad (25.5.11)$$

so Eqs. (25.5.7) and (25.5.3) give

$$\begin{aligned} Q_a \|j, \sigma\rangle^b &= \sum_b e_{ab} Q_b^\dagger \|j, \sigma\rangle \\ &= \sqrt{2M} \sum_b e_{ab} \sum_{\pm} C_{\frac{1}{2}j}(j \pm 1/2, \sigma + b; b, \sigma) \|j \pm 1/2, \sigma + b\rangle. \end{aligned} \quad (25.5.12)$$

From Eqs. (25.5.2), (25.5.3), and (25.5.5) we have

$$Q_a \|j \pm 1/2, \sigma\rangle = \sqrt{2M} C_{\frac{1}{2}j}(j \pm 1/2, \sigma; a, \sigma - a) \|j, \sigma - a\rangle, \quad (25.5.13)$$

while Eqs. (25.5.5), (25.2.31), and (25.5.7) yield

$$Q_a^\dagger \|j \pm 1/2, \sigma\rangle = \sqrt{2M} \sum_b e_{ab} C_{\frac{1}{2}j}(j \pm 1/2, \sigma; b, \sigma - b) \|j, \sigma - b\rangle^b. \quad (25.5.14)$$

Eqs. (25.5.3), (25.5.8), (25.5.10), and (25.5.12)–(25.5.14) give the action of the Q s and $Q^\dagger$ s on all the states of the supermultiplet.

For $j = 0$ we have the collapsed supermultiplet: Eqs. (25.5.3), (25.5.8), (25.5.10), and (25.5.12)–(25.5.14) become

$$\begin{aligned} Q_a \|0, 0\rangle &= 0, & Q_a^\dagger \|0, 0\rangle^b &= 0, \\ Q_a^\dagger \|0, 0\rangle &= \sqrt{2M} \|1/2, a\rangle, & Q_a \|0, 0\rangle^b &= \sqrt{2M} \sum_b e_{ab} \|1/2, b\rangle, \\ Q_a \|1/2, b\rangle &= \sqrt{2M} \delta_{ab} \|0, 0\rangle, & Q_a^\dagger \|1/2, b\rangle &= \sqrt{2M} e_{ab} \|0, 0\rangle^b. \end{aligned} \quad (25.5.15)$$

Now suppose that parity is conserved. Recall that the phase of the supersymmetry generator may be chosen so that the action of the parity operator on these generators is given by Eq. (25.3.13). Then $Q_a^\dagger$ acting on $P\|j, \sigma\rangle$ is a linear combination of the states $PQ_a\|j, \sigma\rangle$, which vanish, and since $P\|j, \sigma\rangle$ has the same rotation properties as $\|j, \sigma\rangle^b$, it must simply be proportional to it

$$P\|j, \sigma\rangle = -\eta \|j, \sigma\rangle^b. \quad (25.5.16)$$

Since P is unitary, η is a phase factor, with $|\eta| = 1$. A corresponding argument shows that $P\|j, \sigma\rangle^b$ is proportional to $\|j, \sigma\rangle$. To find the proportionality coefficient, we note that

$$\begin{aligned} P\|j, \sigma\rangle^b &= (2M)^{-1} P Q_{1/2}^\dagger Q_{-1/2}^\dagger \|j, \sigma\rangle = -\eta (2M)^{-1} Q_{-1/2} Q_{1/2} \|j, \sigma\rangle^b \\ &= -\eta (2M)^{-2} Q_{-1/2} Q_{1/2} Q_{1/2}^\dagger Q_{-1/2}^\dagger \|j, \sigma\rangle = -\eta \|j, \sigma\rangle. \end{aligned}$$

We can then define states of spin j

$$|j, \sigma\rangle^\pm \equiv \frac{1}{\sqrt{2}} \left(|j, \sigma\rangle \pm i |j, \sigma\rangle^b \right), \quad (25.5.17)$$

with definite parity

$$P |j, \sigma\rangle^\pm = \pm i \eta |j, \sigma\rangle^\pm. \quad (25.5.18)$$

Finally, applying the parity operator to Eq. (25.5.5) and using Eqs. (25.3.13) and (25.5.16) gives

$$P |j \pm 1/2, \sigma\rangle = -\frac{\eta}{\sqrt{2M}} \sum_a C_{\frac{1}{2}j}(j \pm 1/2, \sigma; a, \sigma - a) \sum_b e_{ab} Q_b |j, \sigma - a\rangle^b.$$

Eq. (25.5.12) and the orthonormality property of the Clebsch–Gordan coefficients then yield

$$P |j \pm 1/2, \sigma\rangle = \eta |j \pm 1/2, \sigma\rangle, \quad (25.5.19)$$

as was to be shown.

We now turn briefly to the case of extended supersymmetry, with N supersymmetry generators. As mentioned in the previous section, there can be no massless particle with a non-vanishing eigenvalue for any central charge. We can go further and show that the eigenvalues of the central charge operators set a lower bound on the mass of any supermultiplet.

Because the central charges Z_{rs} and $Z_{rs}^\dagger$ commute with each other and with P_μ , one-particle states can be chosen to be eigenstates of all the central charges as well as of P_μ , and because the central charges commute with the Q_{ar} and $Q_{ar}^\dagger$, all states in a supermultiplet have the same eigenvalues.

To derive an inequality relating the mass M of a supermultiplet and the eigenvalues of the central charges on this multiplet, we use the anticommutation relations (25.2.7) and (25.2.8) to write

$$\begin{aligned} \sum_{ar} \left\{ \left(Q_{ar} - \sum_{bs} e_{ab} U_{rs} Q_{bs}^\dagger \right), \left(Q_{ar}^\dagger - \sum_{ct} e_{ac} U_{rt}^* Q_{ct} \right) \right\} \\ = 8NP^0 - 2 \operatorname{Tr} (ZU^\dagger + UZ^\dagger), \end{aligned} \quad (25.5.20)$$

where U_{rs} is an arbitrary $N \cdot N$ unitary matrix. The left-hand side is a positive-definite operator, so by letting this act on the states of the supermultiplet at rest, we find

$$M \geq \frac{1}{4N} \operatorname{Tr} (ZU^\dagger + UZ^\dagger), \quad (25.5.21)$$

where now Z_{rs} denotes the values of the central charges for the supermultiplet of mass M . The polar decomposition theorem tells us that any square matrix Z may be written as HV , where H is a positive Hermitian matrix and V is unitary. We can obtain a useful inequality (which is in fact optimal) by setting $U = V$, in which case Eq. (25.5.21) becomes

$$M \geq \frac{1}{2N} \operatorname{Tr} H = \frac{1}{2N} \operatorname{Tr} \sqrt{Z^\dagger Z}. \quad (25.5.22)$$

States for which M equals the minimum value allowed by this inequality are known as BPS states, by analogy with the Bogomol'nyi–Prasad–Sommerfeld magnetic monopole configurations discussed in Section 23.3, whose mass also equals a lower bound on general monopole masses. In fact, this is more than an analogy; we will see in Section 27.9 that the lower bound on monopole masses in gauge theories with extended supersymmetry is a special case of the lower bound (25.5.22).

As we can see from this derivation of Eq. (25.5.22), for BPS supermultiplets the operator $Q_{ar} - \sum_{bs} e_{ab} U_{rs} Q_{bs}^\dagger$ gives zero when acting on any state of a supermultiplet, so there are only N independent helicity-lowering operators $Q_{(1/2)r}$ and only N independent helicity-raising operators $Q_{(-1/2)r}$, just as in the case of massless supermultiplets. This leads to smaller supermultiplets than would be found in the general case.

For instance, for $N = 2$ supersymmetry the central charge is given by a single complex number²

$$Z = \begin{pmatrix} 0 & Z_{12} \\ -Z_{12} & 0 \end{pmatrix}. \quad (25.5.23)$$

The inequality (25.5.22) here reads

$$M \geq |Z_{12}|/2. \quad (25.5.24)$$

Where $M = |Z_{12}|/2$, the helicity contents of the massive particle supermultiplets are the same as for those of zero mass: there are gauge supermultiplets, consisting of one particle of spin 1, an $SU(2)$ R -symmetry doublet of spin 1/2, and one particle of spin 0 (the other helicity zero state belonging to the spin one particle), and hypermultiplets, consisting of one particle of spin 1/2 and an $SU(2)$ R -symmetry doublet of spin 0. These are sometimes called ‘short’ supermultiplets, to distinguish them from the larger supermultiplets encountered when $M > |Z_{12}|/2$.

²In some articles on $N = 2$ supersymmetry the central charge Z is what we would call $Z/(2\sqrt{2})$.